

MASSENA, NY, USA

WMO: 726223

Lat: 44.936N

Lon: 74.846W

Elev: 65

StdP: 100.54

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 94725

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.2	-22.0	-29.4	0.2	-24.2	-26.4	0.3	-21.2	12.0	-1.9	10.9	-3.4	2.2	220	0.571

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.3	30.8	22.6	29.2	21.6	27.9	20.8	24.0	28.8	22.9	27.4	22.0	26.2	4.9	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.4	17.2	26.9	21.4	16.2	25.7	20.5	15.3	24.9	72.2	28.9	68.2	27.6	64.7	26.3	28.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.4	8.8	8.0	DB	-29.3	33.2	3.9	1.3	-32.1	34.1	-34.4	34.9	-36.6	35.6	-39.4	36.5	(4)
(5)			WB	-29.5	26.0	3.8	1.3	-32.3	26.9	-34.5	27.6	-36.6	28.4	-39.4	29.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.1	-8.9	-7.3	-1.7	6.4	13.5	18.5	21.0	20.0	15.8	9.0	2.4	-4.3	(6)	
	DBStd	11.64	7.71	6.69	6.25	4.86	4.32	3.79	2.95	3.04	4.13	4.62	5.17	6.50	(7)	
	HDD10.0	2333	584	486	364	130	14	0	0	0	3	76	233	443	(8)	
	HDD18.3	4337	843	719	621	360	159	44	8	18	97	289	478	701	(9)	
	CDD10.0	1282	0	0	2	21	124	255	342	309	178	46	4	1	(10)	
	CDD18.3	245	0	0	0	1	11	49	92	69	22	1	0	0	(11)	
	CDH23.3	2234	0	0	3	16	147	463	814	571	207	14	0	0	(12)	
	CDH26.7	584	0	0	0	2	34	125	231	143	49	0	0	0	(13)	
(14) Wind	WSAvg	3.4	3.9	3.9	3.9	4.1	3.5	2.9	2.8	2.6	2.7	3.4	3.6	3.7	(14)	
(15) Precipitation	PrecAvg	877	58	51	59	69	72	79	83	87	90	75	79	76	(15)	
	PrecMax	1203	138	142	118	187	162	159	182	195	182	172	156	170	(16)	
	PrecMin	615	4	8	18	11	27	14	30	20	22	11	29	22	(17)	
	PrecStd	117	31	29	25	37	33	36	39	34	39	34	26	36	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.7	11.2	19.7	25.6	30.1	31.9	32.7	32.2	31.0	25.3	19.1	13.8	(19)	
		MCWB	10.5	7.1	12.8	16.1	20.8	23.4	24.4	22.9	22.5	17.4	13.9	10.1	(20)	
	2%	DB	7.5	6.6	13.4	21.3	27.2	29.8	30.7	29.9	27.9	22.2	16.2	9.4	(21)	
		MCWB	5.7	4.2	8.6	12.9	18.4	21.7	22.7	22.1	21.1	16.6	12.1	7.4	(22)	
	5%	DB	3.9	3.9	9.8	18.0	24.6	27.9	29.1	28.1	25.7	19.2	13.1	6.4	(23)	
		MCWB	2.3	2.1	6.0	10.9	16.6	20.5	21.9	21.1	19.7	14.6	9.6	4.5	(24)	
	10%	DB	1.6	2.2	6.8	15.1	22.2	26.1	27.7	26.6	23.5	17.0	10.3	3.8	(25)	
		MCWB	0.4	0.6	3.9	9.4	15.3	19.3	21.1	20.5	18.6	13.2	7.5	2.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.3	7.9	14.6	17.4	21.9	24.7	25.7	24.7	23.9	19.8	15.1	11.1	(27)	
		MCDB	12.2	10.1	19.0	23.5	28.5	30.5	31.0	29.7	28.7	23.6	17.6	13.1	(28)	
	2%	WB	5.7	4.4	9.5	14.4	20.0	23.0	24.3	23.4	22.0	17.2	12.9	7.6	(29)	
		MCDB	7.3	6.2	12.8	19.2	25.4	28.0	28.8	27.9	26.5	20.6	16.0	9.2	(30)	
	5%	WB	2.5	2.4	6.5	12.0	18.2	21.8	23.0	22.4	20.6	15.2	10.2	4.8	(31)	
		MCDB	3.9	3.8	9.3	16.4	22.5	26.2	27.3	26.4	24.4	18.9	12.8	6.4	(32)	
	10%	WB	0.5	0.7	4.1	9.9	16.5	20.6	22.1	21.3	19.3	13.5	7.8	2.4	(33)	
		MCDB	1.5	1.8	6.5	14.7	20.7	24.5	26.3	25.0	22.8	16.5	10.0	3.7	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.8	10.3	10.0	10.9	12.0	11.4	11.3	11.5	11.9	10.1	8.8	8.2	(35)	
		MCDBR	11.4	10.7	14.1	16.6	15.9	13.8	12.8	13.3	14.2	14.3	13.8	10.1	(36)	
	5% WB	MCWBR	10.4	9.2	10.0	9.9	8.6	7.3	6.6	6.7	8.0	9.2	10.4	8.8	(37)	
		MCWBR	11.2	10.5	13.4	14.0	13.4	11.6	11.2	11.2	12.4	12.8	12.7	9.9	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.270	0.294	0.329	0.354	0.387	0.422	0.434	0.412	0.370	0.343	0.317	0.286	(40)	
		taud	2.397	2.333	2.375	2.395	2.344	2.257	2.226	2.287	2.396	2.471	2.473	2.403	(41)	
	Ebn at Noon	Ebn at Noon	855	892	905	907	882	849	832	838	849	827	790	791	(42)	
		Edh at Noon	73	94	104	112	122	133	136	123	102	83	69	66	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.37	2.29	3.55	4.38	5.14	5.55	5.68	5.03	3.98	2.33	1.45	1.01	(44)	
	RadStd	RadStd	0.16	0.24	0.31	0.51	0.54	0.44	0.48	0.24	0.34	0.23	0.17	0.10	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (35 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page